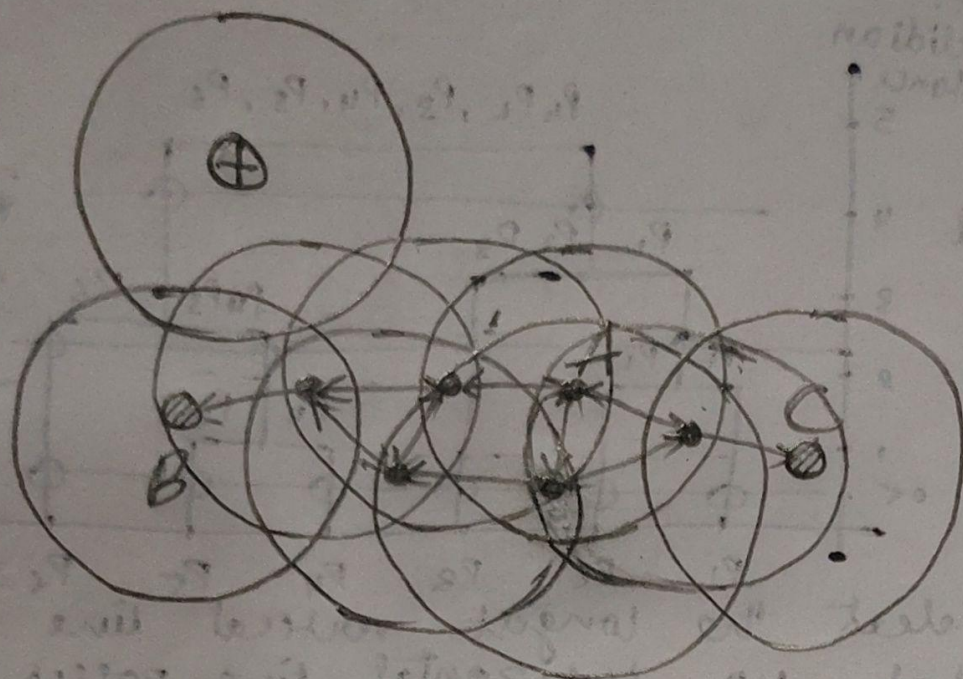


# \* DBSCAN clustering :

M-40

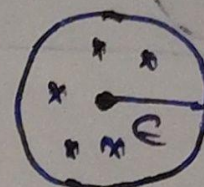


- → core points
  - ⊙ → border points
  - ⊕ → outlier
- Non linear clustering.

$\text{minPts} = 4$

$\epsilon \Rightarrow \text{radius}$

core points



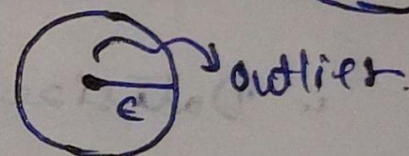
$\text{minPts} = 4$

1. No. of points within the  $\epsilon$  should be greater  $\geq 4$ .

Border point : No. of points within this radius will be less than  $\text{minPts}$ .



Outlier (Noise)



outlier.

In this diagram,  $\text{minPts} = 4$ . Points A and the other red points are core points, because the area surrounding these points in an  $\epsilon$  radius contain at least 4 points (including the point itself). Because they all are reachable from one another, they form a single cluster. Points B and C are not core points, but are reachable from A (via other core points) and thus belong to the cluster as well. Point N is a noise point that is neither a core point nor directly reachable.